

Annarhysa Albert

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PROFESSIONAL SUMMARY

Machine Learning Engineer and AI Engineer pursuing an M.Sc. in Artificial Intelligence Engineering at Universität Passau, with 2+ years of experience across machine learning, NLP, LLMs, data engineering, and software development. Experienced in Python, PyTorch/transformer-based models, LLM fine-tuning, PEFT, ETL pipelines, AWS, Azure, and full-stack development. Experienced in building AI/ML solutions and production data pipelines across research and software engineering environments.

TECHNICAL SKILLS

Programming: Python, Java, C, C++, Go, TypeScript, Shell, MATLAB, SQL, XML, JSON, Excel

AI & Machine Learning: Machine Learning, Deep Learning, NLP, Generative AI, LLMs, LLM Fine-Tuning, Transformers, PEFT, Prompt Tuning, Time-Series Analysis

Data Engineering: ETL, Data Processing, Data Pipelines, SQL, NoSQL, Intelligent Automation

Web & Software Development: Next.js, React.js, Node.js, Vue.js, Spring, Full-Stack Development, REST APIs

Cloud & DevOps: AWS, Azure, Docker, Jenkins, Git, GitHub, Gerrit, JIRA

Languages: English, German, Hindi

PROFESSIONAL EXPERIENCE

Codewalla, Pune (On-site)

Software Development Engineer I

Jul 2025 – Apr 2026

- Engineered an ImageKit wrapper for shisha.com (AIR Global) to cache and manage 50,000+ user-generated images within ETL workflows, improving image delivery and reducing redundant processing.
- Built backend workflows integrating WordPress with serverless AWS functions and UI/UX enhancements, enabling automated data processing and streamlined application workflows.
- Optimised the shisha.com data extraction and management pipeline to process 1000+ records, improving data consistency and reliability across downstream workflows.
- Designed and architected an Azure-based ETL pipeline using Python to automate the extraction, transformation, and processing of 100+ records, reducing manual data-processing effort.
- Implemented full-stack features in Next.js across frontend and backend systems, integrating application components to support end-to-end user workflows.

Nokia, Bangalore (On-site)

SWE Trainee

Sep 2024 – Jun 2025

- Executed 500+ SLNBI and MantaRayNM test scenarios using Robot Framework, NAPET, Python, Shell scripting, and Jenkins, supporting automated performance validation and regression testing.
- Channelized log monitoring with Graylog and remediated 200+ SonarQube code-quality violations, strengthening application observability and QA processes.

HyperVerge, Chennai (On-site)

Nexus Spring '24 ML Trainee

Feb 2024 – Jun 2024

- Constructed a PEFT-based pipeline for fine-tuning Llama 2-7B on Python problem-solving tasks, improving its efficiency by 30%

Bornbhukkad (Remote)

Machine Learning Engineer

Sep 2023 – Apr 2024

- Analyzed 500+ scraped restaurant reviews using transformer-based NLP models and food engineering principles to extract insights from unstructured text, supporting the curation of competitive and health-conscious restaurant menus.

EDUCATION

Master of Science: Artificial Intelligence Engineering

Universität Passau, Passau, Germany

GPA: 2.6/1.0

Achievements: Letter of appreciation for dedicating volunteering hours

Courses: DL for Natural Language, Network Science, Digitisation with AI, DL in Finance, Reproducibility Engg.

Bachelor of Technology: Artificial Intelligence

SRM Institute of Science and Technology, Kattankulathur, India

CGPA: 9.62/10

Achievements: Won Merit-Based Scholarship (AY: 2022-23, 2023-24 and 2024-25).

Courses: OOPs, DSA, DL Techniques, ML, Computer Networks, GenAI and LLMs, Node.js and React.js.

PROJECTS

- **Protego:** Created a crime-awareness dashboard with contextual AI tuning for victim-support recommendations and crime-rate analysis. **Tech:** Python, NLP, LLM Fine-Tuning, Time-Series Analysis, Transformers.
- **MoodMate:** An NLP-based conversational assistant using prompt tuning and LLMs to interpret user input and provide context-aware guidance. **Tech:** Python, LLMs, NLP.
- **CareerWise:** An ML-powered career guidance application to support secondary-level students with career exploration and decision-making. **Tech:** Dart, Python, Machine Learning.
- **Rare Word Summarization :** An NLP-based approach to identify rare words in text and generate summaries. **Tech:** NLTK, TF-IDF.
- **Llama 2-7B Fine-Tuning:** Fine-tuned the Llama 2-7B model for Python problem-solving using parameter-efficient methods. **Tech:** LLMs, PEFT, NLTK.

ACCOMPLISHMENTS

- Microsoft Code; Without Barriers '24 Winner.
- Qualified for the third round of Smart India Hackathon (SIH) '23.
- Published research papers: *Crime Awareness Dashboard with Contextual AI Tuning Engine – Protego* and *Mental Health Support Chatbot*.

LEADERSHIP & VOLUNTEERING

- Supporting AYCD: Ubuntu Mondo in international student engagement, social media management, event coordination, and website migration initiatives.
- Served as a helping hand to Don Bosco, SHP, Pune youth through posters, Informatica, and fundraising events.
- Mentored a tech-driven community - GitHub Community SRM.
- Volunteered as an ML researcher at Maiti Labs to build sustainable models.
- Open-source contributor in Social Winter of Code (SWOC) '24 and Social Summer of Code (SSOC) '24.
- Supported underprivileged education efforts as a Crowdfunding Intern at Muskurahat Foundation.